

DEFINABLE SPACES AND QUOTIENTS

Introduction

Up till now our definable sets were always given as subsets of an ambient space R^m , a very convenient restriction that has served us well. In this final chapter we want to break out of this restricted setting, and consider also, say, projective space, and its “definable” subspaces, more generally, spaces that are not given as subsets of R^m , but locally look like definable subsets of R^m . To stay within the context of “spaces of finite type” we require a covering of the spaces of interest by only finitely many “affine” definable patches. This idea is carried out in detail in Section 1. The main result of this section, theorem (1.8), generalizes a theorem of Robson for semialgebraic spaces, to the effect that a “definable space” (obtained by gluing finitely many affine definable sets) is isomorphic to an affine definable set if and only if the space is **regular**, a separation condition that is easily verified in many situations of interest.

In Section 2 we consider a related construction, namely that of taking the quotient space X/E of a definable set X by a definable equivalence relation E on X . (Definably gluing finitely many definable sets can be viewed as a special case of this construction, but is better treated separately, as we do in Section 1.) We ask when X/E can be realized as an ordinary definable set, and are particularly interested in the case that E is “definably proper” over X . The main positive result in this direction extends a semialgebraic theorem of Brumfiel to our setting. The proof follows Brumfiel’s, with some modifications. Triangulation is the key tool.

The two sections are independent. The point of this chapter is that the entire theory of the previous chapters is not restricted to just definable sets in R^m : we are free to use certain constructions on these definable sets that carry us *outside* R^m .

We fix an o-minimal expansion $\mathcal{R} = (R, <, \dots)$ of an ordered real closed field. “Definable set” will always mean “definable subset of R^m for some m ”, unless specified otherwise.

§1. Definable spaces

(1.1) We start by recalling the construction of a topological space by gluing: Let a covering $S = \bigcup_i U_i$ of a set S by subsets U_i ($i \in I$) be given, and for each index $i \in I$ a set-theoretic bijection $g_i : U_i \rightarrow U'_i$, with U'_i a topological space, such that for all i, j the set $g_i(U_i \cap U_j)$ is open in U'_i and the “transition” map

$$g_{ij} : g_i(U_i \cap U_j) \rightarrow g_j(U_j \cap U_i) \text{ with } g_{ij}(x) = g_j(g_i^{-1}(x))$$

is continuous (hence a homeomorphism, since g_{ij} is a bijection with inverse g_{ji}). Then we equip S with the unique topology in which each U_i is open, and each $g_i : U_i \rightarrow U'_i$ is a homeomorphism. (In this topology a subset X of S is open iff $g_i(X \cap U_i)$ is open for all $i \in I$.)

(1.2) Suppose in addition that the index set I is finite, that each $U'_i \subseteq R^{m(i)}$ is a definable set (with the induced topology from $R^{m(i)}$), and that for each pair i, j the set $g_i(U_i \cap U_j) \subseteq U'_i$ is definable and $g_{ij} : g_i(U_i \cap U_j) \rightarrow g_j(U_j \cap U_i)$ is definable. (Such a family $(g_i : U_i \rightarrow U'_i)_{i \in I}$ is called a **definable atlas on S** , and the g_i ’s **charts** of the atlas.)

Then we extend the notion of “definable set” to subsets of S : Let $X \subseteq S$ and $f : X \rightarrow R$; call X **definable** if $g_i(X \cap U_i) \subseteq U'_i$ is definable for each i , and call f **definable** if X is definable and $f_i : g_i(X \cap U_i) \rightarrow R$ given by $f_i(x) = f(g_i^{-1}(x))$ is definable for each i . (We need this mainly for open X and continuous f .) Note that the definable subsets of S form a boolean algebra of subsets of S , and that for given definable $X \subseteq S$ the definable functions $f : X \rightarrow R$ form an R -algebra under pointwise addition and multiplication of functions. (The constant functions $X \rightarrow R$ are of course definable.) If $X = X_1 \cup \dots \cup X_n$, where all X_k are definable, then a function $f : X \rightarrow R$ is definable iff each restriction $f|_{X_k} : X_k \rightarrow R$ is definable.

Note that U_i is definable, and for $g_i = (g_{i1}, \dots, g_{im(i)}) : U_i \rightarrow R^{m(i)}$, each g_{ik} is definable. Let $\text{DO}(S)$ be the collection of definable open subsets of S , and for each $U \in \text{DO}(S)$, let $\text{DC}(U)$ be the R -algebra of definable continuous functions $f : U \rightarrow R$. Then we call the set S equipped with the sheaf $(\text{DC}(U))_{U \in \text{DO}(S)}$ a **definable space**. This is of course only a sheaf in a *finite* sense: if $U = U_1 \cup \dots \cup U_n$ where $U_k \in \text{DO}(S)$ for all k , then a function $f : U \rightarrow R$ belongs to $\text{DC}(U)$ iff $f|_{U_k} : U_k \rightarrow R$ belongs to $\text{DC}(U_k)$ for all k . The topology on S is not lost in viewing S as a definable space, since $\text{DO}(S)$ is a basis for the topology. However, the particular definable atlas $(g_i : U_i \rightarrow U'_i)_{i \in I}$ that makes S into a definable space is not recoverable from the sheaf, and different definable atlases on S can make S into the same definable space. Two definable atlases $(g_i : U_i \rightarrow U'_i)_{i \in I}$ and $(h_j : V_j \rightarrow V'_j)_{j \in J}$ (on the abstract set S) are called **equivalent**, if for all $i \in I$ and $j \in J$ the sets $g_i(U_i \cap V_j)$ and $h_j(V_j \cap U_i)$ are open definable subsets of U'_i and V'_j respectively, and the transition maps $x \mapsto h_j(g_i^{-1}(x)) : g_i(U_i \cap V_j) \rightarrow h_j(V_j \cap U_i)$ and their inverses $y \mapsto g_i(h_j^{-1}(y))$ are definable and continuous. One checks easily that then (g_i) and (h_j) give rise to the same topology on S , to the same notion of definable subset of S , and to the same notion of definable function on a definable subset of S .

CONCLUSION. Two definable atlases (g_i) and (h_j) on S are equivalent iff they give rise to the same sheaf on S , that is, they make S into the same definable space. The notions of definable subset of S and definable function on a definable subset of S are now seen to depend only on S viewed as a definable space, not on the particular definable atlas used to make S into a definable space.

(1.3) Let T be a second definable space with sheaf $(\text{DC}(V))_{V \in \text{DO}(T)}$. Then a **morphism** $F : S \rightarrow T$ is a map from S into T , such that if $V \in \text{DO}(T)$ and $h \in \text{DC}(V)$, then $F^{-1}(V) \in \text{DO}(S)$ and $h \circ (F|_{F^{-1}(V)}) \in \text{DC}(F^{-1}(V))$. It follows in particular that F is continuous. If the definable space T is determined by the definable atlas $(h_j : V_j \rightarrow V'_j)_{j \in J}$, then a map $F : S \rightarrow T$ is a morphism iff for all $i \in I$ and $j \in J$ the set $g_i(U_i \cap F^{-1}(V_j))$ is definable and open in U'_i , and the map $g_i(U_i \cap F^{-1}(V_j)) \rightarrow V'_j$ given by $x \mapsto h_j(F(g_i^{-1}(x)))$ is continuous and definable. Let $F : S \rightarrow T$ be a morphism. Then, given any definable set $Y \subseteq T$ and definable function $f : Y \rightarrow R$, the set $F^{-1}(Y) \subseteq S$ is definable and the function $f \circ (F|_{F^{-1}(Y)}) : F^{-1}(Y) \rightarrow R$ is definable. Also, if $X \subseteq S$ is definable, then $F(X) \subseteq T$ is definable.

The identity map $S \rightarrow S$ is a morphism, and each constant map $S \rightarrow T$ is a morphism. The definable spaces with their morphisms form a category under the usual composition of maps.

(1.4) EXAMPLES.

(1) Each definable set $S \subseteq R^m$ is made into a definable space by taking the identity map $S \rightarrow S$ as the only chart of a definable atlas. Then the topology of the definable space S equals the topology induced by R^m , and set $X \subseteq S$ is definable in the definable space S iff it is definable as a subset of R^m , and in that case a function $f : X \rightarrow R$ is definable in the sense of the definable space S iff $\Gamma(f)$ is a definable subset of R^{m+1} . If $T \subseteq R^n$ is a second definable set, then a morphism $S \rightarrow T$ is the same thing as a continuous definable map $S \rightarrow T$. An **affine** definable space is by definition a definable space isomorphic to a definable set (in the category of definable spaces with their morphisms).

(2) (Projective spaces) Let $\mathbb{P}^n(R)$ be n -dimensional projective space over the field R ; its points are the equivalence classes $(x_0 : \cdots : x_n)$ of nonzero vectors $(x_0, \dots, x_n)$ in R^{n+1} , where two such vectors are equivalent iff one is a scalar multiple of the other. Clearly $\mathbb{P}^n(R)$ is covered by the $n+1$ subsets U_i ($0 \leq i \leq n$), where U_i is the set of points $(x_0 : \cdots : x_i : \cdots : x_n)$ with $x_i \neq 0$ (or equivalently, with $x_i = 1$). Let $g_i : U_i \rightarrow R^n$ be the bijection $(x_0 : \cdots : 1 : \cdots : x_n) \mapsto (x_0, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$. It is easy to see that $(g_i)_{0 \leq i \leq n}$ is a definable atlas, and we make $\mathbb{P}^n(R)$ into a definable space using this atlas. In fact, $\mathbb{P}^n(R)$ is an affine definable space. To see this, define $v : \mathbb{P}^n(R) \rightarrow R^{(n+1)^2}$ by

$$v(x_0 : x_1 : \cdots : x_n) = (x_i x_j / (x_0^2 + \cdots + x_n^2))_{0 \leq i, j \leq n}.$$

Then v is easily seen to be an isomorphism from the definable space $\mathbb{P}^n(R)$ onto a

definable subset of $R^{(n+1)^2}$.

(3) (Subspaces) Let S be a definable space given by a definable atlas $(g_i : U_i \rightarrow U'_i)_{i \in I}$, and let $X \subseteq S$ be definable. Then $(g_i|_{U_i \cap X} : U_i \cap X \rightarrow g_i(U_i \cap X))_{i \in I}$ is a definable atlas on the set X , and we make X into a definable space using this atlas. We call X in the role of definable space a definable subspace of S . The topology of X as a definable space is the one induced by S . The definable subsets of the definable space X are exactly the definable subsets of S that are contained in X , and given a definable set $Y \subseteq X$, a function $f : Y \rightarrow R$ is definable in the sense of the definable space X iff f is definable in the sense of S . (Thus another (equivalent) definable atlas on S makes X into the same definable space.) The inclusion $X \rightarrow S$ is a morphism. If $F : S \rightarrow T$ is a morphism between definable spaces S and T , and $F(S) \subseteq Y$, where Y is a definable subset of T , then F is also a morphism when considered as a map from S into the definable subspace Y of T .

(4) (Products) Let $S_1, \dots, S_K$ be definable spaces, and take for each S_k a definable atlas $(g_{ki} : U_{ki} \rightarrow U'_{ki})_{i \in I_k}$, $U'_{ki} \subseteq R^{m(ki)}$, $1 \leq k \leq K$.

Put $I := I_1 \times \dots \times I_K$, and for each $i = (i_1, \dots, i_K) \in I$, let

$$\begin{aligned} U_i &:= U_{1i_1} \times \dots \times U_{Ki_K} \subseteq S_1 \times \dots \times S_K, \\ U'_i &:= U'_{1i_1} \times \dots \times U'_{Ki_K} \subseteq R^{m(1i_1) + \dots + m(Ki_K)}, \\ g_i &:= g_{1i_1} \times \dots \times g_{Ki_K} : U_i \rightarrow U'_i. \end{aligned}$$

Then $(g_i : U_i \rightarrow U'_i)_{i \in I}$ is a definable atlas on $S_1 \times \dots \times S_K$ and we make $S_1 \times \dots \times S_K$ into a definable space using this atlas. This gives the product topology on $S_1 \times \dots \times S_K$. Moreover, given a definable space X and morphisms $\mu_k : X \rightarrow S_k$ for $k = 1, \dots, K$, the map $\mu = (\mu_1, \dots, \mu_K) : X \rightarrow S_1 \times \dots \times S_K$ is a morphism. It follows easily that the definable space structure on the product set $S_1 \times \dots \times S_K$ does not depend on our choice of definable atlases for the definable spaces $S_1, \dots, S_K$.

(5) (Definable maps) Let S and T be definable spaces. Then a map $F : S \rightarrow T$ is said to be **definable** if its graph $\Gamma(F)$ is a definable subset of the product space $S \times T$, as specified in (4). One checks easily that for $T = R$ this agrees with the function $F : S \rightarrow R$ being definable in the sense of (1.2). Also $F : S \rightarrow T$ is a morphism iff F is definable and continuous.

(1.5) DEFINITION. Recall that a topological space S is said to be **regular** if for each $a \in S$ and open $U \subseteq S$ with $a \in U$ there is an open $V \subseteq S$ with $a \in V$ and $\text{cl}(V) \subseteq U$. We leave it as an exercise to show that a definable space S is regular if and only if for each $a \in S$ and definable open $U \subseteq S$ with $a \in U$ there is a definable open $V \subseteq S$ with $a \in V$ and $\text{cl}(V) \subseteq U$, and also if and only if for each $a \in S$ and definable closed $X \subseteq S$ with $a \notin X$ there are disjoint definable open neighborhoods of a and X in S .

Note that each affine definable space is regular. The theorem below states the converse. A regular topological space is clearly Hausdorff, but the following is an

example due to Robson of a definable Hausdorff space S that is not regular, and hence not affine.

EXAMPLE. In the o-minimal structure $(\mathbb{R}, <, 0, 1, +, -, \cdot)$, let $U_1 = U'_1 \subseteq \mathbb{R}^2$ be the union of the open unit disc and the point $p := (0, 1)$. Let $U_2 = U'_2 \subseteq \mathbb{R}^2$ be the open square $\{(x, y) : 0 < x < 1 \text{ and } 0 < y < 1\}$. Let $S := U_1 \cup U_2$ as a point set and consider S as a definable space via the identity mappings $g_i : U_i \rightarrow U'_i$ for $i = 1, 2$. With these definitions, S is a definable Hausdorff space. Now consider the arc $\gamma := \{(x, y) : x^2 + y^2 = 1, x > 0, y > 0\} \subseteq U_2$: γ is closed in S since it is closed in U_2 and disjoint from U_1 , while any open subset of U_2 containing γ meets U_1 in an open set whose closure contains p . Thus p and γ cannot be separated.

(1.6) DEFINITION. Let S be a definable Hausdorff space and (a, b) an interval. A definable map $\gamma : (a, b) \rightarrow S$ is called a **definable curve** in S . For $x \in S$, we write $\gamma \rightarrow x$ to mean $\lim_{t \rightarrow b} \gamma(t) = x$. We call γ **completable** if there is a (necessarily unique) point $x \in S$ such that $\gamma \rightarrow x$. If $\gamma \rightarrow x \in X$ with definable $X \subseteq S$ such that $\gamma(a, b) \subseteq X$, we call γ **completable in X** . With these definitions one easily verifies the following facts using (4.1) and (4.2) of Chapter 6:

(1.7) LEMMA. Let S and T be definable Hausdorff spaces, $f : S \rightarrow T$ a definable map, and $x \in S$. Then f is continuous at the point x if and only if for each definable curve γ in S with $\gamma \rightarrow x$ we have $f(\gamma) \rightarrow f(x)$, where $f(\gamma) := f \circ \gamma$.

(1.8) THEOREM. Every regular definable space is affine.

PROOF. Let S be a regular definable space given by the definable atlas $(h_i : U_i \rightarrow V_i)_{i \in I}$ with definable $V_i \subseteq \mathbb{R}^{n_i}$. We clearly can assume each V_i is bounded. Let $l = |I|$. The case $l = 1$ is trivial. If we assume that the theorem holds for $l = 2$, then a straightforward argument proves it for all l .

We now turn to the proof of the case $l = 2$; say $I = \{1, 2\}$. If $X \subseteq \mathbb{R}^n$, we denote by ∂X the set $\text{cl}(X) - X$. Writing $V_{12} := h_1(U_1 \cap U_2)$ and $V_{21} := h_2(U_2 \cap U_1)$, the following definable sets need to be considered:

$$\begin{aligned} B_1 &:= V_1 \cap \partial V_{12} = h_1(\partial U_2), \\ B_2 &:= V_2 \cap \partial V_{21} = h_2(\partial U_1), \\ B'_1 &:= \{x \in \mathbb{R}^{n_1} : \exists y \in B_2 \forall \epsilon_1, \epsilon_2 > 0 \exists z \in U_1 \cap U_2 [d(x, h_1(z)) < \epsilon_1, d(y, h_2(z)) < \epsilon_2]\}, \\ B'_2 &:= \{y \in \mathbb{R}^{n_2} : \exists x \in B_1 \forall \epsilon_1, \epsilon_2 > 0 \exists z \in U_1 \cap U_2 [d(x, h_1(z)) < \epsilon_1, d(y, h_2(z)) < \epsilon_2]\}. \end{aligned}$$

The following fact about these sets is crucial to the construction:

CLAIM 1. $d(x, B'_i) > 0$ for every x in V_i , $i = 1, 2$.

PROOF. We prove the claim for $i = 1$, the case $i = 2$ being symmetrical. Assume for a contradiction that $d(x, B'_1) = 0$ for some $x \in V_1$. Note that $h_1^{-1}(x) \in U_1$ and $U_1 \cap h_2^{-1}(B_2) = \emptyset$, so $U_1 \cap h_1^{-1}(B_2) = \emptyset$. Since S is regular there are disjoint open neighborhoods D of $h_1^{-1}(x)$ in U_1 and E of $h_2^{-1}(B_2)$ in U_2 . We will derive a contradiction by finding an element in $D \cap E$. Since $h_1(D)$ is open in V_1 , we can take $\epsilon > 0$ such that $B(x, \epsilon) \cap V_1 \subseteq h_1(D)$. Because $d(x, B'_1) = 0$ there is $x' \in B'_1$ with $d(x, x') < \epsilon$. By definition of B'_1 there is $y \in B_2$ for which there are points $z \in U_1 \cap U_2$ with $h_1(z)$ arbitrarily close to x' and $h_2(z)$ arbitrarily close to y . Since $h_2(E)$ is an open neighborhood of y in V_2 it follows in particular that there is $z \in U_1 \cap U_2$ with $d(x, h_1(z)) < \epsilon$ and $h_2(z) \in h_2(E)$, hence $z \in D \cap E$. This proves the claim. $\square$

Let $d_1(z) := d(h_1(z), B'_1)$ and $d_2(z) := d(h_2(z), B'_2)$ for $z \in S$. We now define the map $h: S \rightarrow R^{1+n_1+1+n_2}$ as follows:

$$h(z) := \begin{cases} (d_1(z), d_1(z)h_1(z), 0, \dots, 0) & \text{for } z \in U_1 - U_2, \\ (d_1(z), d_1(z)h_1(z), d_2(z), d_2(z)h_2(z)) & \text{for } z \in U_1 \cap U_2, \\ (0, 0, \dots, 0, d_2(z), d_2(z)h_2(z)) & \text{for } z \in U_2 - U_1. \end{cases}$$

By claim 1 we clearly have for each $z \in S$

$$\begin{aligned} h(z) \in U_1 - U_2 &\Leftrightarrow \text{the last } 1 + n_2 \text{ coordinates of } h(z) \text{ are equal to } 0, \\ h(z) \in U_2 - U_1 &\Leftrightarrow \text{the first } 1 + n_1 \text{ coordinates of } h(z) \text{ are equal to } 0. \end{aligned}$$

From this it follows easily, again using claim 1, that h is injective.

CLAIM 2. *The map h is continuous.*

To see this, let γ be a definable curve in S with $\gamma \rightarrow z \in S$. We have to show that $h(\gamma) \rightarrow h(z)$. By restricting the domain of γ suitably we may assume that γ lies either completely in $U_1 - U_2$, or completely in $U_1 \cap U_2$, or completely in $U_2 - U_1$.

CASE 1. γ lies completely in $U_1 - U_2$. Then also $z \in U_1 - U_2$, since $z \in U_2$ would imply that γ lies at least partly in U_2 . The definition of h now gives that $h(\gamma) \rightarrow h(z)$.

CASE 2. γ lies completely in $U_1 \cap U_2$. If also $z \in U_1 \cap U_2$, the definition of h gives $h(\gamma) \rightarrow h(z)$. So let $z \notin U_1 \cap U_2$, say $z \notin U_1$. Then $z \in \partial U_1$, so $y := h_2(z) \in B_2$. Take $x \in R^n$, such that $h_1(\gamma) \rightarrow x$. Since $h_2(\gamma) \rightarrow y \in B_2$ it follows that $x \in B'_1$, hence $h(\gamma) \rightarrow h(z)$ as the formula above shows.

The case that γ lies completely in $U_2 - U_1$ is symmetric to case 1. This finishes the proof of claim 2. $\square$

CLAIM 3. *h maps S homeomorphically onto $h(S)$.*

To prove this, let K be an arbitrary definable closed subset K of S . In view of the injectivity of h and claim 2 it suffices to show that $h(K)$ is then closed in $h(S)$. Let $z \in S$ with $h(z) \in \text{cl}(h(K)) \cap h(S)$; it is enough that we derive from this that $z \in K$. Since $h(z) \in \text{cl}(h(K))$ there is a definable curve γ in K such that $h(\gamma) \rightarrow h(z)$. We may assume that γ lies either completely in $U_1 - U_2$, or completely in $U_1 \cap U_2$, or completely in $U_2 - U_1$. We may also assume the domain of γ is an interval $(0, \epsilon)$.

CASE 1. *The curve γ lies completely in $U_1 - U_2$.* Since

$$h(\gamma(t)) = (d_1(\gamma(t)), d_1(\gamma(t))h_1(\gamma(t)), 0, 0, \dots, 0) \rightarrow h(z) \text{ as } t \rightarrow \epsilon,$$

the last $1 + n_2$ coordinates of $h(z)$ are 0, so that $z \in U_1 - U_2$, and

$$h(z) = (d_1(z), d_1(z)h_1(z), 0, 0, \dots, 0).$$

Using claim 1 this gives $h_1(\gamma) \rightarrow h_1(z)$, hence $\gamma \rightarrow z$, so $z \in K$, as K is closed in S .

CASE 2. *The curve γ lies completely in $U_1 \cap U_2$.* Then $h(\gamma(t)) \rightarrow h(z)$ as $t \rightarrow \epsilon$, where

$$h(\gamma(t)) = (d_1(\gamma(t)), d_1(\gamma(t))h_1(\gamma(t)), d_2(\gamma(t)), d_2(\gamma(t))h_2(\gamma(t))).$$

If $z \in U_1$, then $h(z) = (d_1(z), d_1(z)h_1(z), \dots)$, so that by claim 1 we obtain $h_1(\gamma) \rightarrow h_1(z)$, and thus $\gamma \rightarrow z$ (since h_1 is a homeomorphism onto V_1), hence $z \in K$. If $z \in U_2$ the argument uses instead the last n_2 coordinates of $h(z)$.

The case that γ lies completely in $U_2 - U_1$ is symmetric to case 1. This finishes the proof of claim 3, and thereby the proof of the theorem. $\square$

§2. Definable quotient spaces

(2.1) Given a definable map $f: X \rightarrow Y$ between definable sets X and Y , we denote by E_f the kernel of f , that is, $E_f = \{(x, y) \in X \times X : f(x) = f(y)\}$, a definable equivalence relation on the set X . Note that if f is continuous, then E_f is closed in $X \times X$.

(2.2) Let $E \subseteq X \times X$ be a definable equivalence relation on a definable set $X \subseteq R^m$.

DEFINITION. A **definable quotient of X by E** is a pair (p, Y) consisting of a definable set $Y \subseteq R^n$ and a definable continuous surjective map $p: X \rightarrow Y$ such that:

- (i) $E = E_p$, that is $(x_1, x_2) \in E \Leftrightarrow p(x_1) = p(x_2)$, for all $x_1, x_2 \in X$;
- (ii) p is definably identifying, that is, for all definable $K \subseteq Y$, if $p^{-1}(K)$ is closed in X , then K is closed in Y . (See Chapter 6, (4.4).)

(2.3) REMARKS.

- (1) If clause (ii) above is replaced by the stronger clause

“ p is definably proper”,

then we call (p, Y) a **definably proper quotient of X by E** . (In the presence of surjectivity “definably proper” implies “definably identifying”, see Chapter 6, (4.6).)

- (2) If (p, Y) and (p', Y') are both definable quotients of X by E , then by (i) we clearly have a unique definable bijection $h: Y \rightarrow Y'$ such that the diagram

$$\begin{array}{ccc} & X & \\ p \swarrow & & \searrow p' \\ Y & \xrightarrow[\sim]{h} & Y' \end{array}$$

commutes; by (ii) this bijection is a homeomorphism.

Loosely speaking this means that up to isomorphism there is at most one definable quotient of X by E . If such a quotient (p, Y) exists, we are therefore justified in calling it *the* definable quotient of X by E , and write $Y = X/E$. We shall also use the phrase “ $p: X \rightarrow Y$ is a definable quotient of X by E ” instead of “ (p, Y) is a definable quotient of X by E ”.

- (3) A definable quotient $p: X \rightarrow Y$ is a quotient of X by E in the category of definable sets and continuous definable maps, in the sense that if $f: X \rightarrow Z$ is any continuous definable map into a definable set $Z \subseteq \mathbb{R}^n$ such that $E \subseteq E_f$, then the unique map $g: Y \rightarrow Z$ such that $f = g \circ p$ is continuous and (obviously) definable.
- (4) For a definable quotient of X by E to exist it is clearly necessary that E is closed in $X \times X$. But this is by no means sufficient, see Brumfiel [7, p. 71].

(2.4) EXAMPLE. Let $X \subseteq \mathbb{R}^m$ be definable and $A \subseteq X$ a nonempty definable subset of X . We would like to “definably collapse” A to a point. In precise terms, consider the definable equivalence relation E_A on X whose equivalence classes are the singletons $\{x\}$ with $x \in X - A$, and the set A ; we would like to find the definable quotient X/E_A . Concerning this we have the following result.

PROPOSITION. *If A is closed and bounded in $\mathbb{R}^m$, then there is a definable proper quotient of X by E_A .*

PROOF. Consider the definable continuous map $p: X \rightarrow \mathbb{R}^{m+1}$ given by $p(x) = (d_A(x) \cdot x, d_A(x))$, where $d_A(x) := \inf\{|x - a| : a \in A\}$ is the distance from x to A . Put $Y := p(X)$, and consider p as a map $X \rightarrow Y$. Assume A is closed and bounded. Claim: (p, Y) is a definably proper quotient of X by E_A . Clause (i) is clearly satisfied. To check the “proper” version of (ii), consider a definable curve $\gamma: (b, c) \rightarrow X$ such that $p(\gamma) \rightarrow p(x) \in Y$, $x \in X$. By Chapter 6, (4.5) it suffices to

show that γ is completable in X . If $\lim_{t \rightarrow c} d_A(\gamma(t)) = 0$, then $\gamma \rightarrow a$ for some $a \in A$, since A is closed and bounded in R^m , and we are done. So we may assume that $\lim_{t \rightarrow c} d_A(\gamma(t)) = d_A(x) \neq 0$. In combination with $\lim_{t \rightarrow c} d_A(\gamma(t)) \cdot \gamma(t) = d_A(x) \cdot x$, this gives $\lim_{t \rightarrow c} \gamma(t) = x$, and we are done again. $\square$

This is a very special case of our main theorem (2.15), but before we can get to that we need a number of preliminaries, on completions, disjoint sums, and attaching spaces respectively.

(2.5) COMPLETIONS.

A **completion** of a definable set $S \subseteq R^m$ is a pair (h, S') consisting of a closed and bounded definable set $S' \subseteq R^n$ (for some n) and a definable map $h: S \rightarrow S'$ such that h is a homeomorphism from S onto $h(S)$ and $h(S)$ is dense in S' . We also express this more informally by: $h: S \rightarrow S'$ is a completion of S .

Clearly each definable set $S \subseteq R^m$ has a completion: just take a definable map $\mu: R^m \rightarrow R^m$ that maps R^m homeomorphically onto $(-1, 1)^m$, put $S' := \text{closure of } \mu(S)$, and let $h: S \rightarrow S'$ be the restriction of μ . Then $h: S \rightarrow S'$ is a completion of S .

(2.6) Let $f: S \rightarrow T$ be a definable continuous map between definable sets $S \subseteq R^m$ and $T \subseteq R^n$. Then a **completion of $f: S \rightarrow T$** is a triple consisting of a completion $i: S \rightarrow S'$ of S , a completion $j: T \rightarrow T'$, and a definable continuous map $f': S' \rightarrow T'$ such that

$$f' \circ i = j \circ f.$$

We also express this by saying that the commutative diagram

$$(*) \quad \begin{array}{ccc} S & \xrightarrow{i} & S' \\ f \downarrow & & \downarrow f' \\ T & \xrightarrow{j} & T' \end{array}$$

is a **completion diagram** of $f: S \rightarrow T$.

There is always a completion of $f: S \rightarrow T$, but for our purpose we need a more precise result:

(2.7) LEMMA. *Let a completion $j: T \rightarrow T'$ be given. Then there is a completion diagram $(*)$ of $f: S \rightarrow T$ with the given map j as the bottom map in $(*)$. If in addition S is given as a definable closed subset of a definable set $X \subseteq R^m$, then there are a completion $h: X \rightarrow X'$ and a completion diagram $(*)$ of f , such that $S' = \text{closure of } h(S)$ in X' , and $i = h|_S: S \rightarrow S'$.*

PROOF. Choose a completion $g: S \rightarrow S^C$ of S , and note that the definable map $g \times j: S \times T \rightarrow S^C \times T'$ is a homeomorphism onto its image $g(S) \times j(T)$. Let

S' be the closure of $(g \times j)(\Gamma(f))$ in $S^C \times T'$. Then $i: S \rightarrow S'$ given by $i(x) = (g \times j)(x, f(x)) = (g(x), jf(x))$ is a completion of S , and the restriction of the projection map $S^C \times T' \rightarrow T'$ to S' is a definable continuous map $f': S' \rightarrow T'$ making the diagram (*) commutative.

Suppose now that S is a definable closed subset of the definable set $X \subseteq R^m$. In the ambient affine space of T' we take a closed ball B that contains T' . By Chapter 8, (3.10) there is a definable continuous extension $\tilde{f}: X \rightarrow B$ of $j \circ f: S \rightarrow T'$. Apply the first part of the lemma to get a completion of $\tilde{f}: X \rightarrow B$ consisting of a completion $h: X \rightarrow X'$, the completion $1_B: B \rightarrow B$, and a definable continuous map $\tilde{f}': X' \rightarrow B$ such that $\tilde{f}' \circ h = \tilde{f}$. Put $S' := \text{closure of } h(S) \text{ in } X'$, $i := h|_S: S \rightarrow S'$, and note that $\tilde{f}'(S') \subseteq T'$. Hence $f' := \tilde{f}'|_{S'}$ provides us with a completion diagram (*) as required. $\square$

(2.8) DISJOINT SUMS.

Let $S_1, \dots, S_k$ be definable sets in $R^{m(1)}, \dots, R^{m(k)}$, $k \geq 1$.

DEFINITION. A **disjoint sum** of $S_1, \dots, S_k$ is a tuple $(h_1, \dots, h_k, T)$ consisting of a definable set $T \subseteq R^n$ for some n , and definable maps $h_i: S_i \rightarrow T$ such that:

- (i) h_i is a homeomorphism onto $h_i(S_i)$ and $h_i(S_i)$ is open in T , for $i = 1, \dots, k$;
- (ii) T is the disjoint union of the sets $h_1(S_1), \dots, h_k(S_k)$.

Let $n = 1 + \max\{m(i) : 1 \leq i \leq k\}$ and define $h_i: S_i \rightarrow R^n$ by $h_i(x) := (x, i, \dots, i)$. Then $(h_1, \dots, h_k, \bigcup_i h_i(S_i))$ is clearly a disjoint sum of $S_1, \dots, S_k$.

If $(h_1, \dots, h_k, T)$ and $(h'_1, \dots, h'_k, T')$ are both disjoint sums of $S_1, \dots, S_k$, then there is a unique definable homeomorphism $\lambda: T \rightarrow T'$ such that $\lambda \circ h_i = h'_i$ for $i = 1, \dots, k$. This means, loosely speaking, that there is up to isomorphism exactly one disjoint sum of $S_1, \dots, S_k$. In the following we fix such a disjoint sum $(h_1, \dots, h_k, T)$ of $S_1, \dots, S_k$, and we write $S_1 \amalg \dots \amalg S_k$ for T ; there will be no harm in identifying each S_i with its image in $S_1 \amalg \dots \amalg S_k$ via h_i .

(2.9) CONSTRUCTION OF $X \amalg_f Y$.

Let $X \subseteq R^m$ and $Y \subseteq R^n$ be definable sets, and let a definable continuous map $f: A \rightarrow Y$ from a definable set $A \subseteq X$ into Y be given. We would like to attach X to Y via f . To this end we consider on the disjoint sum $X \amalg Y$ the smallest equivalence relation $E(f)$ for which each $a \in A \subseteq X$ is equivalent to $f(a) \in Y$. One checks easily that

$$E(f) = \Delta(X) \cup \Delta(Y) \cup \{(a, f(a)) : a \in A\} \cup \{(f(a), a) : a \in A\} \\ \cup \{(a_1, a_2) \in A \times A : f(a_1) = f(a_2)\},$$

where $\Delta(X) = \{(x, x) : x \in X\}$ and $\Delta(Y) = \{(y, y) : y \in Y\}$ are the diagonals of X and Y . This shows that $E(f)$ is a *definable* equivalence relation; the question is

whether the definable quotient of $X \amalg Y$ by $E(f)$ exists. If it does, we denote this quotient by $X \amalg_f Y$ rather than by $X \amalg Y/E(f)$, and we say that $X \amalg_f Y$ is the **space obtained by attaching X to Y via f** .

(2.10) UNIVERSAL PROPERTY OF $X \amalg_f Y$.

Suppose $X \amalg_f Y$ exists. Let $\xi: X \rightarrow X \amalg_f Y$ be the composition

$$X \rightarrow X \amalg Y \rightarrow X \amalg_f Y,$$

and similarly for $\eta: Y \rightarrow X \amalg_f Y$. Clearly ξ and η are definable and continuous and $\xi(a) = \eta(f(a))$ for all $a \in A$. Moreover, given any two definable continuous maps $\xi': X \rightarrow Z$ and $\eta': Y \rightarrow Z$ into a definable set Z with $\xi'(a) = \eta'(f(a))$ for all $a \in A$, there is exactly one map $\mu: X \amalg_f Y \rightarrow Z$ such that $\mu \circ \xi = \xi'$ and $\mu \circ \eta = \eta'$; this map μ is continuous and (obviously) definable.

(2.11) LEMMA. *Let A be closed and bounded in the ambient space R^m of X . Then $X \amalg_f Y$ exists as a definably proper quotient of $X \amalg Y$.*

PROOF. If $A = \emptyset$, then the identity map $X \amalg Y \rightarrow X \amalg_f Y$ is clearly a definably proper quotient of $X \amalg Y$ by $E(f)$. So let us assume $A \neq \emptyset$. Let R^M be the ambient space of $X \amalg Y$. We identify X and Y with their images in $X \amalg Y$; note that A is then closed and bounded in R^M . Let $d_A: R^M \rightarrow R$ be the distance function given by

$$d_A(z) := \min\{|z - a| : a \in A\}.$$

Let also $\tilde{f}: X \rightarrow R^M$ be a continuous definable extension of $f: A \rightarrow Y$. (Such $\tilde{f}$ exists by Chapter 8, (3.10).) Define the map $p: X \amalg Y \rightarrow R^{2M+1}$ by

$$p(x) = (\tilde{f}(x), d_A(x) \cdot x, d_A(x)) \text{ for } x \in X \text{ and } p(y) = (y, 0, 0) \text{ for } y \in Y,$$

so p is definable and continuous, and once checks easily that $E_p = E(f)$. Let $Z = p(X \amalg Y)$, a subset of R^{2M+1} . We will show that (p, Z) is the desired definably proper quotient of $X \amalg Y$ by $E(f)$. Let γ be a definable curve γ in $X \amalg Y$ such that $p(\gamma)$ is completable in Z .

By Chapter 6, (4.5) it suffices to show that γ is completable in $X \amalg Y$. We may as well assume (after restricting the domain of γ suitably) that either γ lies entirely in X , or γ lies entirely in Y . In the first case we may further assume γ lies entirely in $X - A$ and is bounded away from A since otherwise γ will be completable in X with $\gamma \rightarrow a \in A$; with this further assumption on γ and using the assumption that $p(\gamma)$ is completable in Z it follows just as in the proof of (2.4) that γ is completable in X . The case that γ lies entirely in Y is even simpler and left to the reader. $\square$

We now use completions to extend this lemma as follows:

(2.12) PROPOSITION. *Suppose A is closed in X and $f: A \rightarrow Y$ is definably proper. Then $X \amalg_f Y$ exists as a definably proper quotient of $X \amalg Y$.*

PROOF. Identifying X and Y with their images in suitable completions and using lemma (2.7) we may assume that X and Y are bounded in their ambient space and f extends to a definable continuous map $\text{cl}(f): \text{cl}(A) \rightarrow \text{cl}(Y)$. Since f is definably proper we have $\text{cl}(f)^{-1}(Y) = A$. (To see this, let $x \in \text{cl}(A)$, $\text{cl}(f)(x) \in Y$. To derive $x \in A$ we take a definable path $\gamma: [0, 1] \rightarrow \text{cl}(A)$ with $\gamma(1) = x$ and $\gamma([0, 1)) \subseteq A$. Then $\text{cl}(f)(\gamma([0, 1))) \subseteq Y$ is closed and bounded in the ambient affine space of Y , so $f^{-1}(\text{cl}(f)(\gamma([0, 1))))$ is a subset of A and is closed and bounded in the ambient affine space of X . Since this set contains $\gamma([0, 1))$ it must also contain $\gamma(1) = x$, hence $x \in A$.) By (2.11) we have a definably proper quotient $\text{cl}(p): \text{cl}(X) \amalg \text{cl}(Y) \rightarrow X \amalg_{\text{cl}(f)} Y$. Consider $X \amalg Y$ as a subset of $\text{cl}(X) \amalg \text{cl}(Y)$ in the obvious way and let $Z := \text{cl}(p)(X \amalg Y)$. It is easily seen that $\text{cl}(p)^{-1}(Z) = X \amalg Y$ and that $E(\text{cl}(f)) \cap (X \amalg Y)^2 = E(f)$. Hence $p := \text{cl}(p)|_{X \amalg Y}: X \amalg Y \rightarrow Z$ is a definably proper quotient of $X \amalg Y$ by $E(f)$. $\square$

(2.13) Let E be a definable equivalence relation on a definable set X . Let $pr_1: E \rightarrow X$ and $pr_2: E \rightarrow X$ be the restrictions of the two projection maps $X \times X \rightarrow X$.

We call E **definably proper over X** if pr_1 (equivalently, pr_2) is a definably proper map. One checks that if X/E exists as a definably proper quotient of X by E , then E is definably proper over X . The main theorem (2.15) below asserts the converse. We need one more lemma before embarking on its proof.

(2.14) LEMMA. *Let $E \subseteq X \times X$ be a definable equivalence relation on a definable set $X \subseteq R^m$ and suppose E is definably proper over X . Then, given a definable curve $\gamma = (\alpha, \beta)$ in E with $\alpha = pr_1(\gamma)$, $\beta = pr_2(\gamma)$, if either α or β is completable in X , both are completable in X , and in that case, if $\alpha \rightarrow p$, $\beta \rightarrow q$, then $\gamma \rightarrow (p, q) \in E$. (Hence E is closed in $X \times X$.)*

PROOF. Suppose α is completable in X . Since $pr_1: E \rightarrow X$ is definably proper and γ lifts α , it follows that γ is completable in E . Hence $\beta = pr_2(\gamma)$ is completable in X . $\square$

(2.15) THEOREM. *Suppose the definable equivalence relation E on the definable set X is definably proper over X . Then X/E exists as a definably proper quotient of X .*

This contains proposition (2.12) on $X \amalg_f Y$ as a special case, but (2.12) is needed in the proof of (2.15). Here are two further important special cases:

(2.16) COROLLARY. *If $X \subseteq R^m$ is closed and bounded and $E \subseteq X \times X$ is a closed definable equivalence relation, then X/E exists as a definably proper quotient of X .*

(2.17) APPLICATION TO ORBIT SPACES. Let G be a definable topological group, that is, G is a definable subset of some R^m equipped with a definable continuous group operation $G \times G \rightarrow G$. (We leave it as an exercise to show that then the group inversion $x \mapsto x^{-1} : G \rightarrow G$ is also continuous, so that G is indeed a topological group.) Let in addition a continuous definable group action $G \times X \rightarrow X$ on a definable set X be given. Then the orbits Gx are exactly the equivalence classes of the definable equivalence relation $\sim_G$ on X given by $x \sim_G y \Leftrightarrow gx = y$ for some $g \in G$.

(2.18) COROLLARY. *If G is closed and bounded (for example, finite), then the orbit space $X/G := X/\sim_G$ exists as a definably proper quotient of X .*

(2.19) PROOF OF THEOREM (2.15). By induction on $\dim(X)$. If $\dim(X) \leq 0$, then X is finite; in that case the theorem holds trivially. Assume $\dim(X) = d > 0$. By Chapter 6, (1.2), there is a definable set $S \subseteq X$ that has exactly one point in common with each equivalence class of E . Let $\sigma : X \rightarrow S$ be the definable map that assigns to each $x \in X$ the unique point in S to which it is equivalent; σ may not be continuous, let alone definably proper. Nevertheless we can use σ and S to construct X/E . Let $c(S)$ be the closure of S in X . Let $B = \sigma(c(S) - S)$ be the set of points of S that are equivalent to points of $c(S) - S$, so $\dim(B) < d$. Take a triangulation of X that is compatible with S , $c(S)$ and B , let $\mathcal{P}$ be the partition of X corresponding to this triangulation, and let S_d be the union of the d -dimensional sets of $\mathcal{P}$ that are contained in S . One checks easily that S_d is open in X . Put $S' := c(S) - S_d$. Since $B \cap S_d = \emptyset$, no point of S_d is equivalent to a point of S' . Moreover S' is closed in X . Hence $E' := E \cap (S' \times S')$ is definably proper over S' . Since $\dim(S') < d$ there is by the inductive hypothesis a definably proper map $f' : S' \rightarrow Y'$ onto a definable set Y' with $E' = E_{f'}$. Note that f' maps $S - S_d$ bijectively onto Y' . Let $c(S_d)$ be the closure of S_d in X , and put $A := c(S_d) \cap S'$. We construct $Y = X/E$ by attaching $c(S_d)$ to Y' via $f'' := f'|_A : A \rightarrow Y'$. Note that A is closed in $c(S_d)$, and that $f'' : A \rightarrow Y'$ is definably proper, since f' is. Hence we can apply proposition (2.12) to obtain $Y := c(S_d) \amalg_{f''} Y'$ as a definably proper quotient of $c(S_d) \amalg Y'$ via the map $p : c(S_d) \amalg Y' \rightarrow Y$.

Note that the composed map $S' \rightarrow Y' \rightarrow Y$ agrees with the map $c(S_d) \rightarrow Y$ on the intersection A of their domains, hence these two maps determine a (definable, continuous) map $g : c(S) = c(S_d) \cup S' \rightarrow Y$. Consider the following commuting diagram:

$$\begin{array}{ccc} c(S_d) \amalg S' & \xrightarrow{j} & c(S_d) \amalg Y' \\ h \downarrow & & \downarrow p \\ c(S) & \xrightarrow{g} & Y = c(S_d) \amalg_{f''} Y' \end{array}$$

where the (continuous, definable) map j is the identity on $c(S_d)$ and equal to f' on S' , and the (continuous, definable) map h is induced by the inclusion maps $c(S_d) \rightarrow c(S)$ and $S' \rightarrow c(S)$. Note that all four maps are surjective, that p is definably proper, and that j is definably proper since f' is. An easy diagram chase then shows that g is definably proper. Since f' maps $S - S_d$ bijectively onto Y' ,

the map $g|_S: S \rightarrow Y$ is a bijection. Now put $f := g \circ \sigma: X \rightarrow S \rightarrow Y$. Clearly f is definable, surjective, and $E = E_f$. As mentioned already, σ may not be continuous, but we claim

(*) $f: X \rightarrow Y$ is continuous and definably proper.

(Clearly (*) implies that $f: X \rightarrow Y$ is a definably proper quotient of X by E , so we have reduced to proving (*).) To check continuity of f , take a definable curve α in X with $\alpha \rightarrow p \in X$. Then $(\alpha, \sigma(\alpha))$ is a definable curve in E and α is completable in X , hence $\sigma(\alpha)$ is completable in X by lemma (2.14), say $\sigma(\alpha) \rightarrow q \in c(S)$. Then $(p, q) \in E$, and $f(\alpha) = g(\sigma(\alpha)) \rightarrow g(q)$. But $g(q) = f(q)$: if $q \in S$, this follows from $\sigma(q) = q$; if $q \in c(S) - S$, then $\sigma(q) \in S - S_d$, and both q and $\sigma(q)$ are in S' , so that $f'(q) = f'(\sigma(q))$, hence $g(q) = g(\sigma(q)) = f(q)$. Also $f(q) = f(p)$. Hence $f(\alpha) \rightarrow f(p)$. This proves continuity of f . To show f is definably proper, take a definable curve α in X such that $f(\alpha) = g(\sigma(\alpha))$ is completable in Y . Since g is definably proper, $\sigma(\alpha)$ is completable in $c(S)$. Again by lemma (2.14) it follows that α is completable in X . $\square$

Notes and comments

The notion of definable space (even for arbitrary o-minimal structures) is implicit in Pillay [48]. It is used there to study the properties of groups and fields that are definable in o-minimal structures. See also Khovanskii [33] for similar gluing constructions in the Pfaffian setting.

For the proof of theorem (1.8) I largely followed Robson's treatment in [50] of the semialgebraic case. The material on completions in Section 2 is adapted from Delfs and Knebusch [12], and the remainder of Section 2 follows Brumfiel [8].

For o-minimal expansions of "the real field with restricted analytic functions" there is also a quite different way to go beyond the affine setting, by introducing a (locally defined) notion of "nice" subset of an arbitrary real analytic manifolds, generalizing "subanalytic" sets; see [23].